Product design, Plant location, Choice of markets/sources, Production structure, Distribution/Dealer network design, Location of Warehouses, Plant Layout and Logistics, Allocation Design, Production Planning, Inventory Management – Stocking Levels, Transportation-mode Choice, Shipment Size and Routing Decisions, and Transport Contracting, Packaging, Materials Handling, Warehousing Operations.

D. Key Actors

Shippers (users of logistics), Suppliers (of logistics services) – Carriers (rail, road, air, water, pipeline, rope-way), Warehouse Providers, Freight Forwarders, Terminal Operators (ports, stevedores etc), Government (regulator of logistics).

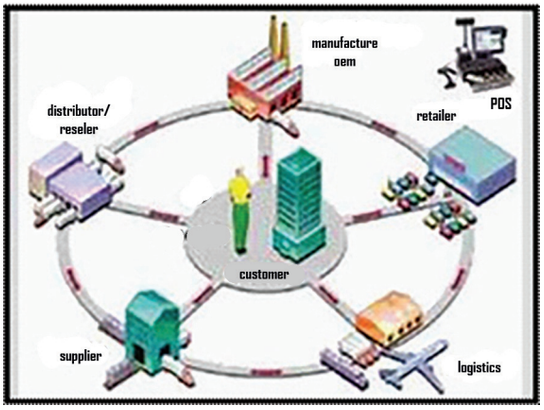
Organisations taking proactive managerial attention in co ordinating the actors in logistics leads to reduced logistics costs and improved customer service.

E . Role of Government

The government plays a significant role in logistics in India. The important legislations that affect logistics are Central Sales Tax and Local Sales Tax, Consignment Tax, Excise Duties, Octroi and Entry Tax, Use of Packaging Material, MODVAT, GST, Motor Vehicles Act and similar acts for other modes, Distribution of Policies.

F. Classification of Logistics Applications

The Logistics Management can be classified on the basis of applications from various dimensions in the process of examining and evaluating alternatives. They are 1. Decision-wise 2. Actor-wise 3.Inbound logistics and Outbound logistics





G. Elements of Logistics Cost

The important elements of logistics cost are

Product Inventory at source, Pipeline Inventory, Product Inventory at Warehouses and Dealers, Transit Losses/Insurance, Storage Losses/Insurance, Handling and Warehouse Operations, Packaging, Transportation, Customers' Shopping.

H. Models in Logistics Management

The decision areas of Logistics can be addressed using various quantitative models from Operations Research namely

i. Forecasting Models ii. Mathematical Programming Models – Location Models, Allocation Models, Distribution Network Design Models iii. Inventory Models iv. Routing Models v. Scheduling Models vi. Alternatives Analysis

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